

# Helix Thready — Aurora OS (Qt / Silica)

## Customization Spec

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| Rev | Date       | Author                     | Change                                                                                                                                                                                                                                                                                                                                                              |
|-----|------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | 2026-07-22 | swarm (design · platforms) | Initial consolidation of the scattered per-screen Aurora one-liners into one design contract: Silica page-stack navigation, remorse-timer pattern, ambience-vs-token tension (honest), Qt Accessibility mapping, component substitutions, density buckets (THREADY-DES-05 inherited), motion honesty, keyboard/hardware back; minted<br>THREADY-DES-PLAT-05/-06/-07 |

**Honest status header.** The Aurora client is a `helix_shims` **skeleton** [GAP: 8.5] , and its would-be token package ( `helix_design` Qt arm) is a **verified empty scaffold** [GAP: 8.2/8.3] (platform-map §2). The QT column of the platform matrix has **zero VERIFIED cells** (platform-map §5). Nothing in this file is build-ready or claimed to work; it is the **design contract the client builds against**, and every Silica/Qt API name is **ASSUMED** until verified against current Aurora OS SDK docs and the `helix_shims` contract — [OPEN: THREADY-DES-PLAT-05] (companion to the inherited [OPEN: THREADY-DES-LIB-03] ).

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### 1. Position & sources consolidated

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Aurora guidance previously lived as one-liners in: the 11 mobile artifacts' notes panels (SilicaListView/pulley/remorse/ambience mentions), wireframes §6.1 (Qt Accessibility / ambience / Qt font scaling row), platform-map §3 QT column + §4, brand-assets §5.1 + icon-export-matrix §3.8 (density-bucket PNGs), and design-system §7. This file consolidates and extends them; the sources stay authoritative for their own scope. Typography (Qt font database bundling — ASSUMED; system-default fallback; scale mapping) is in [typography-substitution.md](#).

### 2. Navigation idiom (5-tab IA → Silica page stack)

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Silica's native idiom is a **page stack** (push/pop, edge-swipe back) plus **pulley menus** — not a bottom tab bar. The wireframes §6 IA [\[VERIFIED structure\]](#) is five bottom tabs. The resolution, already sketched by the home-feed artifact note ("page stack + bottom tab strip") and kept honest here:

- **Tab layer** [\[DEFAULT – adjustable\]](#): a custom bottom tab strip renders the five-tab IA for cross-platform parity (same IA everywhere — wireframes §6). This deliberately *deviates from pure Silica idiom*; the alternative (replatforming tabs onto pulley/attached pages) would fragment the parity contract. The trade-off is recorded, not hidden — revisit inside [\[OPEN: TREADY-DES-PLAT-05\]](#) with Aurora HIG evidence.
- **Stack layer** (ASSUMED APIs): within a tab, Silica [PageStack](#) push/pop; edge-swipe back pops (the shared back pattern — home-feed note [\[VERIFIED intent\]](#)).
- **Pulley menus** host per-screen secondary actions, consolidated from the artifact notes: Channels — Add-channel + filters in the pulley; threads — Re-sync/Pause/Recipes; Search — scope switches. Primary actions stay on-surface (parity rule: a pulley may *duplicate*, never *hide*, a load-bearing action [\[DEFAULT – adjustable\]](#)).

- **Sheets:** Add-Channel is a Silica `Dialog` page pushed from the pulley (add-channel note), with the \$3.4 wizard data-preservation rule intact (dismiss never loses input).

Per-screen consolidation (each cell ASSUMED, `[GAP: 8.5]`):

| Screen             | Qt/Silica realization (from the notes panels)                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------|
| Home feed          | Qt Quick/Silica page stack + bottom tab strip; ambience-tint policy per \$4                                |
| Channels           | <code>SilicaListView</code> ; Add-channel + filters via pulley; edge-swipe back                            |
| Add-Channel        | Silica <code>Dialog</code> page pushed from the pulley menu                                                |
| Channel threads    | <code>SilicaListView</code> + pulley (Re-sync/Pause/Recipes); edge-swipe back                              |
| Post detail        | pipeline rows with Qt Accessibility; <b>remorse timer</b> on Reprocess (\$3)                               |
| Search             | Silica <code>SearchField</code> header item over the list                                                  |
| Assets grid        | <code>SilicaGridView</code> ; filters via pulley; Qt font scaling per wireframes \$6.1                     |
| Media viewer       | Qt Multimedia ( <code>MediaPlayer</code> / <code>VideoOutput</code> ) under Silica chrome; edge-swipe back |
| Account / Settings | identical card stack; ambience-aware theme row (\$4); role-gated Admin card                                |
| Notifications      | Aurora events-view integration ( <code>[OPEN: THREADY-DES-15]</code> surface)                              |

### 3. Remorse-timer pattern for destructive actions

**Verified precedent:** the post-detail artifact’s Aurora note specifies a “remorse-timer pattern for Reprocess confirm” `[VERIFIED – screens/mobile/post-detail.html notes panel]`. This file promotes it from a one-liner to the platform rule `[DEFAULT – adjustable]`:

- On Aurora, destructive or hard-to-undo actions use the Sailfish/Aurora **remorse** pattern — an inline countdown (“Reprocessing in 5s — tap to cancel”) that executes on expiry — *instead of* the modal confirm dialog other platforms use. Same decision semantics, platform-native delivery; an allowed deviation under the parity contract because states and outcomes are identical.
- Scope: Reprocess (verified precedent), Remove channel, Pause channel bulk actions, Sign out. **Not** used where the web contract mandates stronger friction (typed confirmation), if any such flow lands — remorse never *weakens* a confirm contract.

- Semantics: the countdown row is announced via Qt Accessibility (state + remaining action); cancel is a full-row tap target ( $\geq 44\text{px}$ ); the destructive label uses `--danger` — remorse styling never masks the semantic color (design-system §6.2 rule).
- Timer length  $\sim 5\text{ s}$  `[DEFAULT — adjustable]`; the `RemorsePopup` /remorse-item API availability in current Aurora Silica is **ASSUMED** — `[OPEN: THREADY-DES-PLAT-05]`. Fallback if absent: the standard confirm dialog (platform-map §3 `Dialog` row).
- Idempotency guard: expiry fires the same idempotent action the other platforms use (e.g. Reprocess 409 “already running” handling — mobile README §3), so a double-fire is harmless.

## 4. Ambience-aware theming vs. the Thready tokens

**The tension, documented honestly.** Silica **ambiences** derive an accent/tint from the user-chosen ambience (wallpaper), i.e. *the OS wants to color your app*. Thready mandates the opposite: every surface is colored by the **token palette** (design-system §3) plus per-Account white-label (theming §8), with semantic colors untouchable. Both cannot fully win. Ground truth already contains the arbitration seed: wireframes §6.1 pins the Aurora theme source as “bridge + Silica ambience”, and the settings artifact note says “ambience-aware; **explicit choice still wins** (parity contract)” `[VERIFIED intents]`.

Resolution `[DEFAULT — adjustable]`, final policy `[OPEN: THREADY-DES-PLAT-07]`:

1. **Tokens always win for load-bearing UI:** text, controls, focus, chips, progress, and every **semantic** color (`--danger` / `--warn` / `--success`) come from the token bridge. A white-label Account brand must survive any ambience; an error must survive both (theming §10.2 rule).
2. **Ambience may tint decorative background layers only** — the page background wash behind token-colored surfaces — so the app still feels ambience-native without breaking AA or brand. If contrast of any token pairing over an ambience-tinted layer cannot be guaranteed, the layer falls back to token `--bg` (AA is non-negotiable, design-system §6.1).
3. **Explicit theme choice wins over ambience-implied lightness:** the Light/System/Dark three-state behaves as on every platform; “System” maps to the ambience’s light/dark character (ASSUMED API for reading it).

## 5. Qt Accessibility mapping

The design-system §6 contract on Qt Accessibility (wireframes §6.1 row `[VERIFIED table]`; all attribute names ASSUMED):

| Contract point                | Qt/Aurora realization                                                                                                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WCAG 2.2 AA contrast          | Carried by tokens; §4.2 guards ambience layers                                                                                                                                                                                                             |
| Brand ≠ text; danger ≠ brand  | Token discipline; remorse rows keep <code>--danger</code> (§3)                                                                                                                                                                                             |
| Focus visible everywhere      | <code>Accessible.focused</code> /focus highlight on every interactive; visible focus for hardware-keyboard use (§9)                                                                                                                                        |
| Keyboard + screen reader      | <code>Accessible.role</code> / <code>.name</code> / <code>.description</code> on every interactive; pipeline rows expose per-step status (“Qt Accessibility on pipeline rows” — post-detail note <code>[VERIFIED intent]</code> ); state changes announced |
| Reduced motion                | System animation-reduction (ASSUMED availability) gates §8; poster-frame freeze                                                                                                                                                                            |
| Localized, direction-agnostic | ru / sr-Cyrl / en; string keys shared with the fleet ( <code>I18nService</code> dictionary, incl. <code>footer.made</code> / <code>footer.by</code> / <code>ally.love</code> <code>[VERIFIED keys]</code> )                                                |

Screen-reader maturity on Aurora is **not asserted** — actual capability on target devices is part of `[OPEN: THREADY-DES-PLAT-05]` verification; the design contract stands regardless.

## 6. Component substitutions (Qt/Silica)

Derived from platform-map §3 (QT column) + §4, upgraded with the Silica-specific counterparts the mobile artifact notes verified as design intent. **Every row ASSUMED** ( `[GAP: 8.5]` ):

| Component group           | Qt/Silica substitution                                                                                                                                                                                                   | Source                                 |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Buttons                   | <code>Button</code> + QML token styles; danger palette for destructive; loading = <code>enabled:false</code> + <code>BusyIndicator</code>                                                                                | platform-map §3                        |
| Input / field             | <code>TextField</code> + label/error <code>Text</code> triple (border tint, message, accessible state)                                                                                                                   | platform-map §3/§4                     |
| Select / date             | <code>ComboBox</code> / <code>Calendar</code> popup — native pickers win on mobile (platform-map §4 rule)                                                                                                                | platform-map §3                        |
| Checkbox / radio / switch | <code>CheckBox</code> / <code>RadioButton</code> / <code>Switch</code> , accent track via token bridge                                                                                                                   | platform-map §3/§4                     |
| Card / stat               | <code>Frame</code> /QML rect composites                                                                                                                                                                                  | platform-map §3                        |
| List / table              | <code>SilicaListView</code> (upgraded from generic <code>TableView</code> for phone surfaces — table collapses to thread-row cards per platform-map §4); <code>TableView</code> only on genuinely tabular admin surfaces | notes panels + §4                      |
| Grid                      | <code>SilicaGridView</code> (assets masonry approximation)                                                                                                                                                               | assets-grid note                       |
| Search                    | <b>Silica</b> <code>SearchField</code> header item                                                                                                                                                                       | search note                            |
| Badge / chips             | pill <code>Rectangle</code> ; hashtag pill + outline (indirect = dimmed + non-color marker); processing chip = pill + <code>BusyIndicator</code> , five states + focusable retry                                         | platform-map §3 + component-library §7 |
| Dialog                    | <code>Dialog</code> (Silica page-style) — but destructive confirms prefer the remorse pattern (§3)                                                                                                                       | platform-map §3 + §3 above             |
| Toast                     | overlay <code>Popup</code> ; never the only channel for a critical error                                                                                                                                                 | platform-map §3/§4                     |
| Progress / spinner        | <code>ProgressBar</code> / <code>BusyIndicator</code> ; helix-motif via QML <code>Canvas</code> / <code>Shape</code> , frozen under reduced motion                                                                       | platform-map §3/§4                     |
| Tooltip                   | <code>ToolTip</code> on long-press; load-bearing hints also visible helper text                                                                                                                                          | platform-map §4                        |
| Skeleton                  |                                                                                                                                                                                                                          | platform-map §3/§4                     |

| Component group         | Qt/Silica substitution                                                                | Source            |
|-------------------------|---------------------------------------------------------------------------------------|-------------------|
|                         | animated rects, ≥150 ms delay before showing                                          |                   |
| Empty / error           | centered composites + retry                                                           | platform-map §3   |
| Media                   | Qt Multimedia <code>MediaPlayer</code> / <code>VideoOutput</code> under Silica chrome | media-viewer note |
| Theme toggle / language | palette swap three-state (§4) / <code>ComboBox</code> locale                          | platform-map §3   |

## 7. Density buckets

Inherited, unresolved: Aurora icon PNGs are produced at **86 / 108 / 128 / 172 / 250 px** ( `[VERIFIED — generate-raster.sh run; icon-export-matrix $3.8]` ), but the bucket set itself is `[RESEARCH]` and must be re-verified against current Aurora packaging docs — `[OPEN: THREASY-DES-05]` **inherited, not re-litigated here**. Layout consequence `[DEFAULT — adjustable]`: express paddings/targets via Silica `Theme` units (ASSUMED: `Theme.paddingSmall/Medium/Large`, `Theme.itemSizeSmall/...`) mapped from the nearest token spacing steps, rather than raw px, so density scaling is the OS's job; the mapping table is produced with the token bridge ( `[OPEN: THREASY-DES-LIB-04]` ).

## 8. Motion

Checked against ground truth: `motion.md` §6 names `lottie-web` / `lottie-compose` / `lottie-ios` — **no Qt or Aurora Lottie runtime is named anywhere** in `motion.md` or `platform-map`. Honest consequence:

- **Lottie on Aurora is unverified and not designed-in.** Candidate routes (rlottie-based QML players, Qt add-ons) are **ASSUMED to exist at best** — evaluation is `[OPEN: THREASY-DES-PLAT-06]`; `THREASY-MOT-01`'s verify-before-ship rule applies.
- **Guaranteed floor:** Qt Quick animations ( `NumberAnimation`, `Behavior`, states/transitions) with the token durations `--motion-fast 150ms` / `--motion-base 200ms` on `--ease-standard cubic-bezier(0.2,0,0,1)` `[VERIFIED values — design-system §5]`, plus the five delivered static poster SVGs (`motion` §5 `[VERIFIED]`) for loading/success/error/pulse/sync states. The helix spinner may alternatively be a QML `Canvas` rotation of the static mark.
- Reduced motion freezes to the exact §5 poster frames (equivalent-runtime rule); remorse countdowns (§3) remain **textual** under reduced motion (the timer is meaning, not decoration — it keeps counting, only decorative easing is dropped).

## 9. Keyboard & hardware back

- **Edge-swipe back is the primary back gesture** (= the shared back pattern, home-feed note `[VERIFIED intent]`); the back-stack rules of `harmonyos.md` §3 apply identically (sheet-dismiss preserves input;

back never confirms a destructive action). Aurora devices with a hardware back key (ASSUMED to exist on some targets) map it to the same pop.

- **Hardware keyboard** (attached/BT) `[DEFAULT — adjustable]`: adopt the verified web keyboard model (wireframes §1.2) additively — `/` or `Ctrl+K` focus search, `Esc` = pop/dismiss (equivalent of edge-swipe), `Tab` / `Enter` / `Space` focus + activate. Same rule as desktop (desktop README §3.1): **never rebind a verified web binding**; platform accelerators are additive. No global system-wide shortcut is claimed.

## 10. Open items

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- `[OPEN: THREADY-DES-PLAT-05]` — **verify every Silica/Qt mapping in this file** (PageStack, pulley API, `RemorsePopup` / remorse item, `SilicaListView` / `SilicaGridView` / `SearchField`, Qt Multimedia on Aurora, ambience-read API, `Theme` units, screen-reader maturity, hardware back-key existence) against the current Aurora OS SDK/HIG and the `helix_shims` contract. Companion to the inherited `[OPEN: THREADY-DES-LIB-03]`.
  - `[OPEN: THREADY-DES-PLAT-06]` — **Aurora motion runtime**: evaluate/verify any Lottie-capable QML player vs. committing to the §8 Qt-native floor; until then only the floor may be claimed.
  - `[OPEN: THREADY-DES-PLAT-07]` — **ambience-vs-token policy**: ratify (or amend) the §4 arbitration — tokens win for load-bearing UI, ambience tints decorative layers only, explicit choice wins — with AA evidence over real ambiences.
  - Inherited: `[OPEN: THREADY-DES-05]` (density buckets `[RESEARCH]`), `[GAP: 8.2/8.3/8.5]` (empty `helix_design` Qt arm; client skeleton — release gate, wireframes §6.2), `[OPEN: THREADY-DES-LIB-03/-04]`, `[OPEN: THREADY-DES-15]` (notifications surface), `[OPEN: THREADY-DES-PLAT-01/-02/-08]` (typography, via `typography-substitution.md`).
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